SUPPLY CHAIN VISIBILITY & OPTIMISATION

Milestone 3 — Performance Analytics

Supplier Scorecards • Transportation Cost
• Route & Carrier Performance • Advanced KPIs

Infosys Springboard Virtual Internship 7.0 — Batch 2

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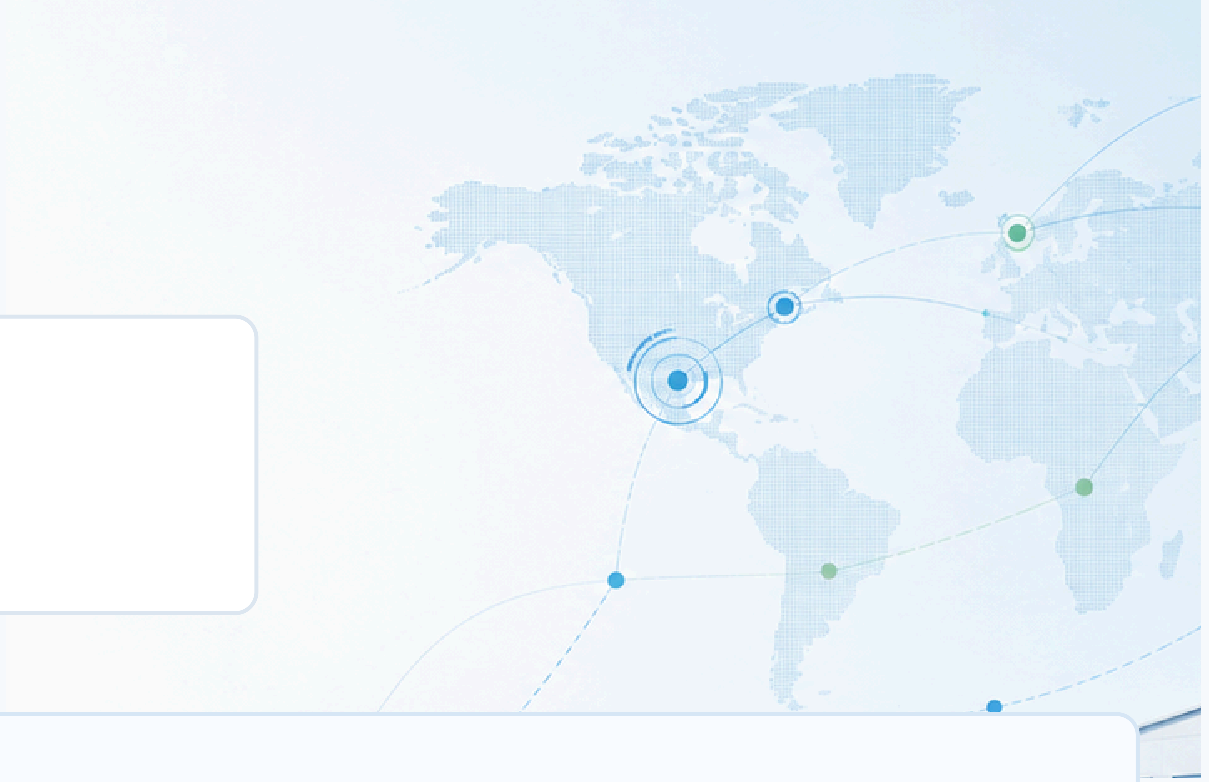
From cleaned data to operational decision support

PROJECT CONTEXT

Project Overview

A unified view of product movement, delivery reliability, freight, sellers and route performance.



Cleaned Records	Distinct Orders	Movement Lines	On-Time Delivery
66,039	55,649	63,089	93.91%
Final analysis-ready dataset	Validated order grain	Unique order-item movements	Overall service reliability

BUSINESS OBJECTIVE

Turn transactional e-commerce data into a practical supply-chain control tower that helps management answer:

- How efficiently are products moving through the network?
- Where are delays, freight costs and regional bottlenecks concentrated?
- Which sellers and routes are strong performers — and which need intervention?
- How can operational KPIs be combined into decision-ready management views?

DATA COVERAGE

Orders	lifecycle & status	**Products**	category & dimensions
Sellers	location & performance	**Customers**	region & city
Freight	cost & efficiency	**Reviews**	experience proxy
Payments	transaction mix	**Delivery**	estimated vs actual

PROGRESSION

Project Journey: Milestone 1 → 2 → 3

Each milestone builds on the previous one — from clean data, to visibility, to performance-oriented analytics.

1

Milestone 1

Data Preprocessing

Cleaned and validated the merged Olist dataset using Python, missing-value treatment, feature engineering and IQR outlier handling.

OUTPUT: 66,039 clean rows

2

Milestone 2

Power BI Visibility

Built a five-page Supply Chain Control Tower for product movement, delivery, regional drilldown, trends and variance.

OUTPUT: 5 Decision-Ready Pages

3

Milestone 3

Performance Analytics

Added seller scorecards, transportation-cost analysis, route performance and advanced cross-KPI management views.

OUTPUT: 4 Analytical Workstreams

The analytical grain and data-quality decisions from Milestone 1 remain the foundation for every KPI that follows.

PREVIOUS MILESTONE

Milestone 1 Recap — Data Preprocessing

Python preprocessing transformed the raw merged Olist data into an analysis-ready dataset.

Raw Records	Exact Duplicates	Before IQR	Cleaned Output
119,143	**0**	**81,170**	**66,039**
40 raw columns	No exact duplicate rows	Rows before outlier filtering	36 final columns

CLEANING WORKFLOW

01	**Profile**	shape • dtypes • nulls • duplicates
02	**Clean**	drop noise • fill categorical/numeric gaps
03	**Validate**	price • freight • product dimensions
04	**Engineer**	delivery_delay from actual vs estimated date
05	**Treat outliers**	IQR-based filtering of extreme values
06	**Export**	Supply_Chain_Cleaned_Dataset.csv

FEATURE ENGINEERING — PYTHON LOGIC

```
delivery_delay =
  delivered_customer_date
  − estimated_delivery_date
```

OUTLIER RULE — IQR

```
IQR = Q3 − Q1
Lower = Q1 − 1.5 × IQR
Upper = Q3 + 1.5 × IQR

Result: 66,039 rows retained
```

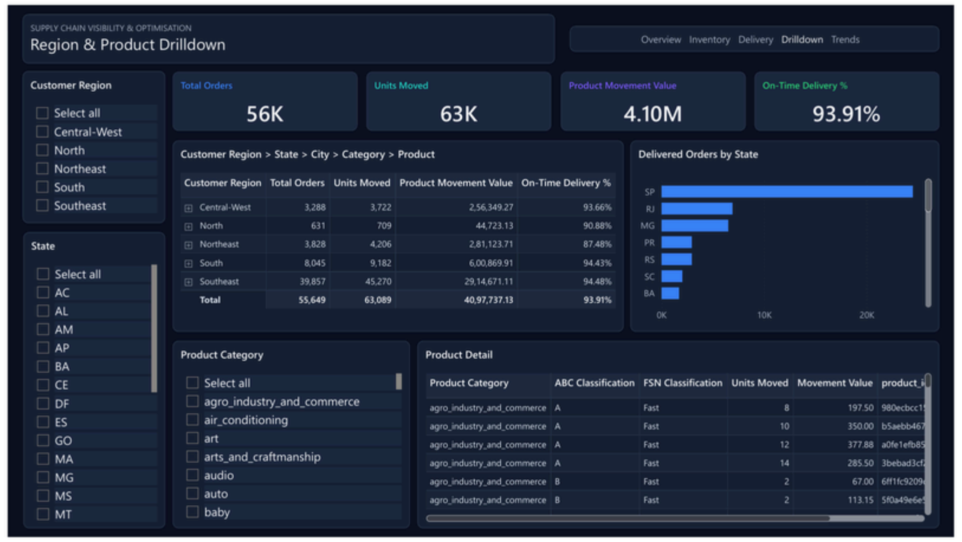
Milestone 2 Recap — Supply Chain Control Tower

Five Power BI pages converted the cleaned dataset into interactive visibility and diagnostic analytics.

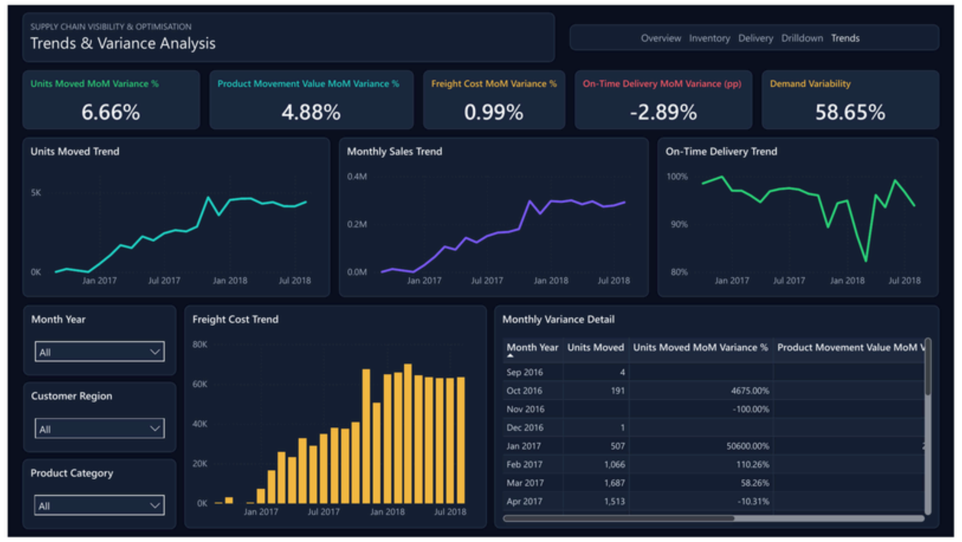
1. Executive Supply Chain Overview

2. Inventory Movement Intelligence

3. Delivery Performance



4. Region & Product Drilldown



5. Trends & Variance Analysis

Milestone 2 Analytical Foundation

The dashboard logic was designed around correct grain, transparent inventory assumptions and duplicate-safe measures.

A Correct analytical grain

63,089 unique Movement Line Keys prevent repeated payment rows from inflating order-item analysis.

B Inventory proxy — not stock

Olist has no warehouse stock or on-hand balance. Product movement is explicitly used as the inventory proxy.

C Service reliability

Delivery KPIs compare actual delivery with estimated delivery and distinguish on-time/early from late orders.

DAX — DUPLICATE-SAFE TOTAL SALES

```
Total Sales =
SUMX(
  VALUES([Movement Line Key]),
  CALCULATE(MAX([price]))
)
```

KPI LOGIC — MOVEMENT & DELIVERY

```
Units Moved = DISTINCTCOUNT([Movement Line Key])

On-Time Delivery % =
  On-Time Delivered Orders / Delivered Orders

Demand Variability = STDEV(monthly movement) / AVERAGE(monthly movement)
```

Why it matters: business insights are only trustworthy when the data grain and measure definitions are correct.

MILESTONE 3

Performance Analytics

Four analytical workstreams deepen the control tower from visibility into performance evaluation, cost diagnosis and decision support.

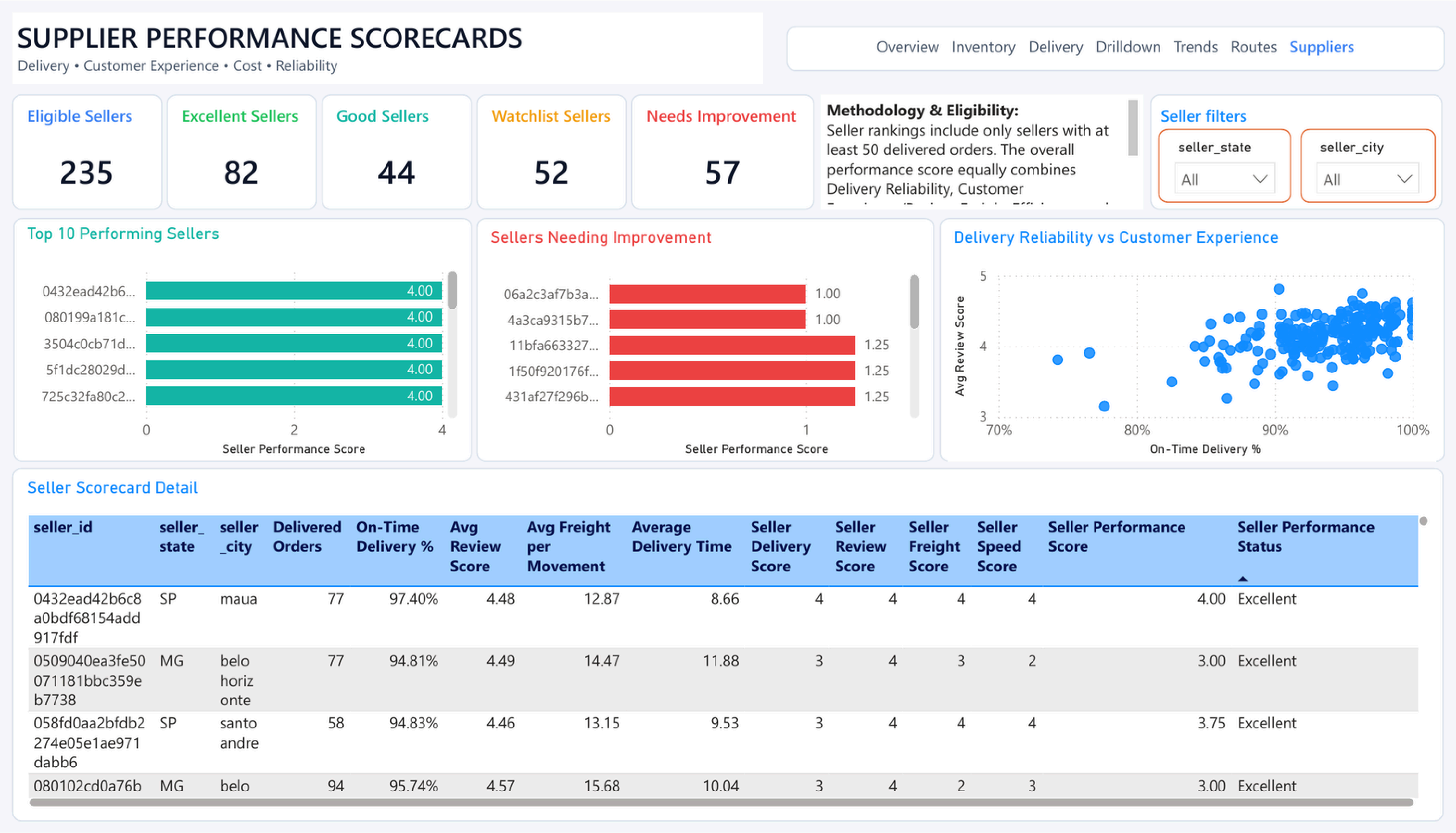
- 01 **Supplier Scorecards**
- 02 **Transportation Cost**
- 03 **Route & Carrier Performance**
- 04 **Advanced KPI Visualization**

Milestone 3 asks not only "what happened?" — but "where is performance strong, weak, costly or inefficient?"

DASHBOARD 1

Supplier / Seller Performance Scorecards

A quartile-based scorecard identifies strong sellers and sellers that require operational attention.



seller_id	seller_state	seller_city	Delivered Orders	On-Time Delivery %	Avg Review Score	Avg Freight per Movement	Average Delivery Time	Seller Delivery Score	Seller Review Score	Seller Freight Score	Seller Speed Score	Seller Performance Score	Seller Performance Status
0432ead42b6c8a0bdf68154add917fdf	SP	maua	77	97.40%	4.48	12.87	8.66	4	4	4	4	4.00	Excellent
0509040ea3fe50071181bbc359eb7738	MG	belo horizonte	77	94.81%	4.49	14.47	11.88	3	4	3	2	3.00	Excellent
058fd0aa2bfdb2274e05e1ae971dabb6	SP	santo andre	58	94.83%	4.46	13.15	9.53	3	4	4	4	3.75	Excellent
080102cd0a76b	MG	belo	94	95.74%	4.57	15.68	10.04	3	4	2	3	3.00	Excellent

Eligible
235
50+ delivered orders

Excellent
82
Top-performing

Watchlist
52
Needs monitoring

Needs Improvement
57
Priority attention

METHODOLOGY

- Formal ranking includes sellers with at least 50 delivered orders.
- Four equally weighted dimensions: Delivery, Review, Freight, Speed.
- Review score is a customer-experience / quality proxy — not direct product quality.
- Performance bands: Excellent • Good • Watchlist • Needs Improvement.

KEY FINDINGS

Supplier Performance — Findings, Scoring Logic & Interpretation

The scorecard combines service, experience, cost and speed into one defensible seller-level performance view.

Performance distribution — 235 eligible sellers

- Excellent 82
- Good 44
- Watchlist 52
- Needs Improvement 57

DAX — OVERALL SELLER PERFORMANCE SCORE

```
Seller Performance Score =
DIVIDE(
  [Seller Delivery Score] + [Seller Review Score] +
  [Seller Freight Score] + [Seller Speed Score],
  4
)

Eligibility: [Seller Delivered Orders] >= 50
```

1 A sizeable improvement pool

109 sellers are either Watchlist or Needs Improvement — 46.4% of the eligible base. Performance governance should focus here first.

2 High review alone is not enough

The scorecard prevents one strong metric from masking weak freight efficiency or slow delivery by balancing all four dimensions equally.

3 Quartiles make the method explainable

Scores from 1–4 are based on the eligible-seller distribution, making the ranking relative, transparent and easy to defend in review.

Overall score bands: Excellent ≥ 3.00 • Good ≥ 2.50 • Watchlist ≥ 2.00 • Needs Improvement < 2.00

DASHBOARD 2

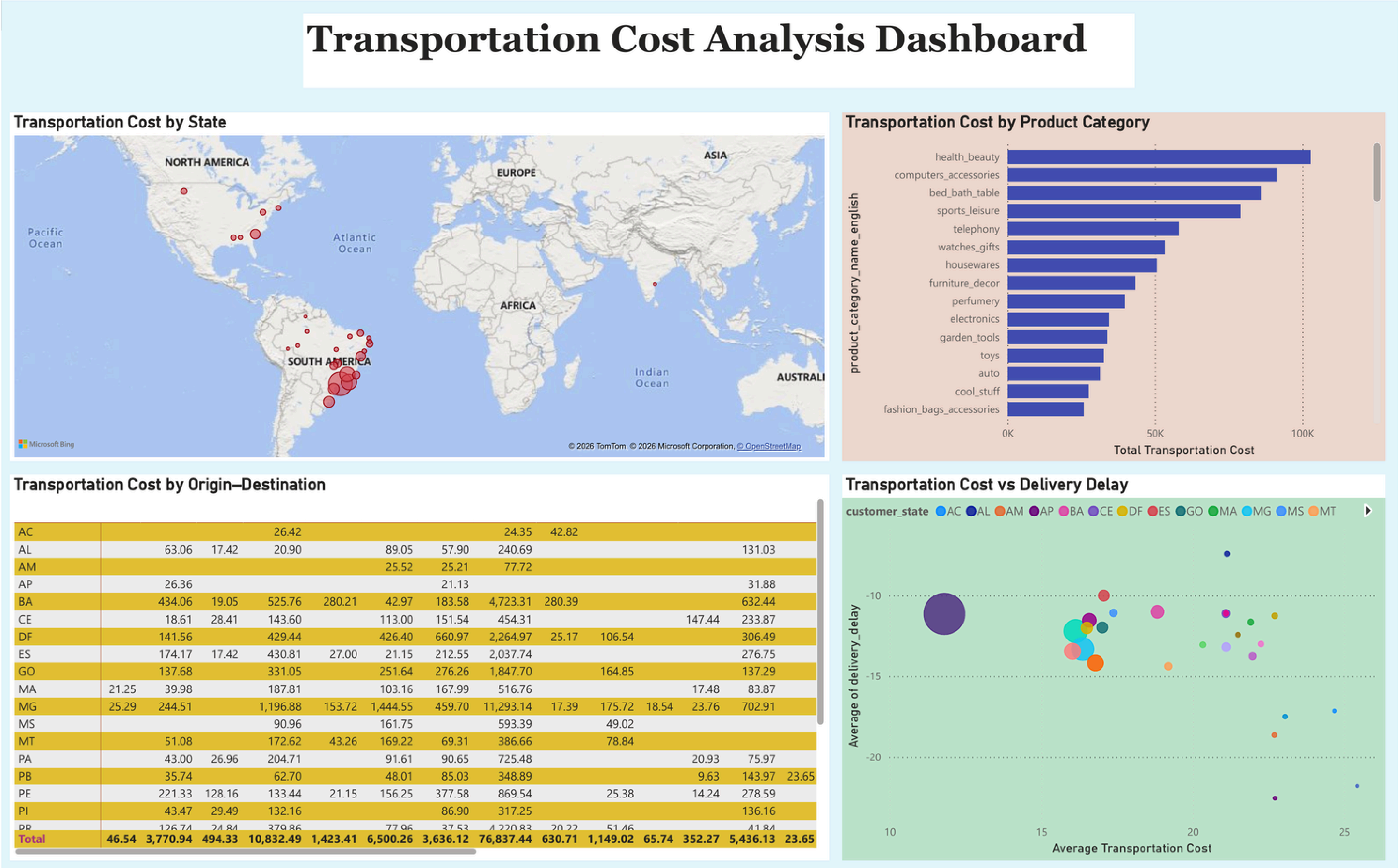
Transportation Cost Analysis — Where Are We Spending?

Freight expenditure is examined by geography, origin–destination movement, product category and delivery outcome.

Total Transportation Cost

969.23K

Dashboard total



1

Geographic concentration

RJ and MG appear among the largest cost contributors in the state-level view.

2

Category concentration

Health & beauty and computer accessories lead transportation cost by product category.

3

Cost ≠ inefficiency by itself

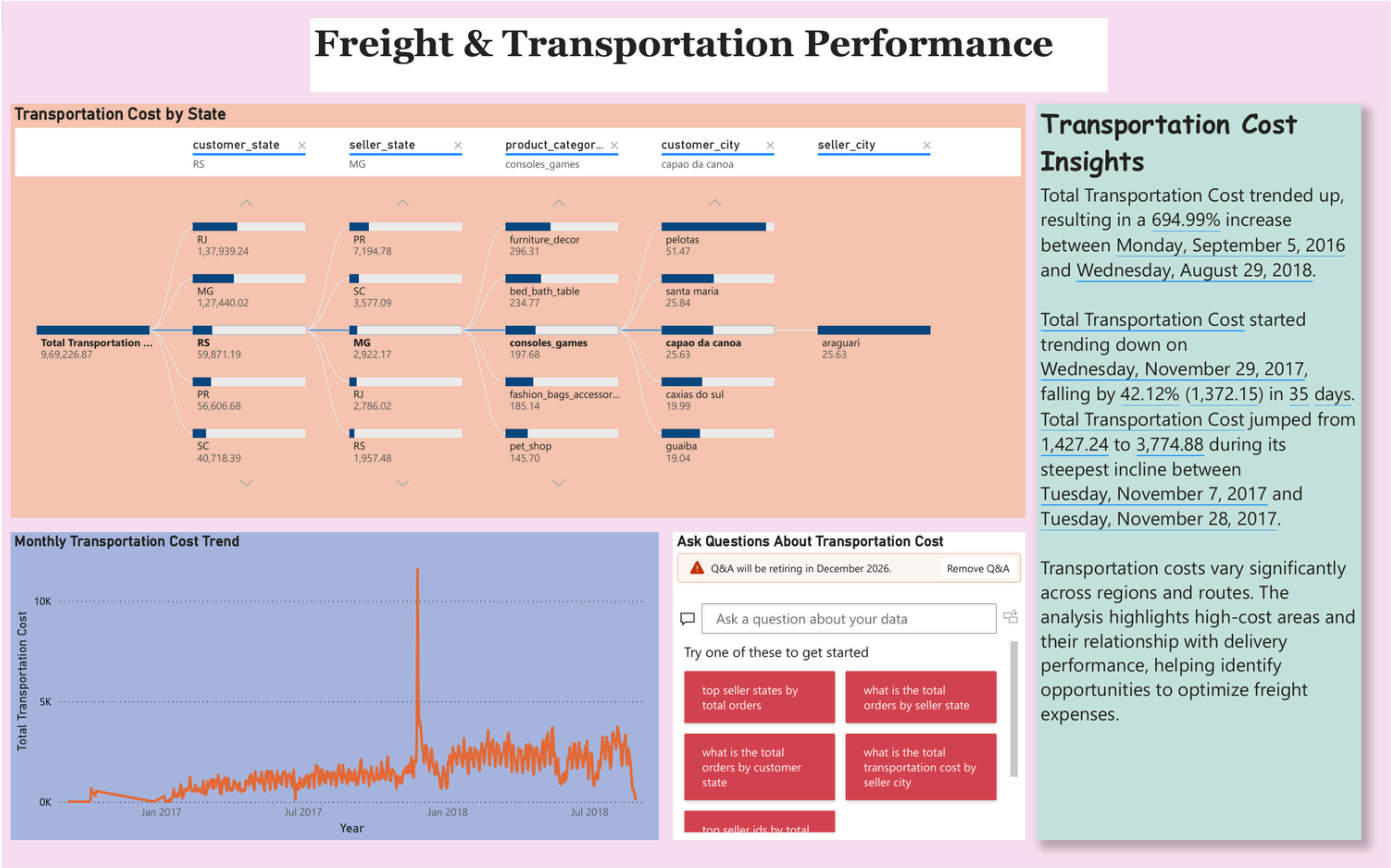
The scatter compares transportation cost with delivery delay so expensive movements can be judged alongside service outcomes.

CORE CALCULATION LOGIC

DASHBOARD 2 — DETAIL

Freight & Transportation Performance — Trends & Drilldown

The second transportation view adds monthly trend analysis, hierarchical drilldown and Power BI-generated narrative insights.



POWER BI INSIGHT

Transportation cost trended upward across the dataset period, with the dashboard insight reporting a 694.99% increase from Sep 2016 to Aug 2018.

- The steepest incline occurred during Nov 2017 in the automatically generated trend insight.
- Cost varies materially across regions, cities and origin–destination combinations.
- Filters allow the analyst to isolate specific seller/customer geography and categories.
- The monthly trend reveals spikes that deserve root-cause analysis instead of relying on the grand total.

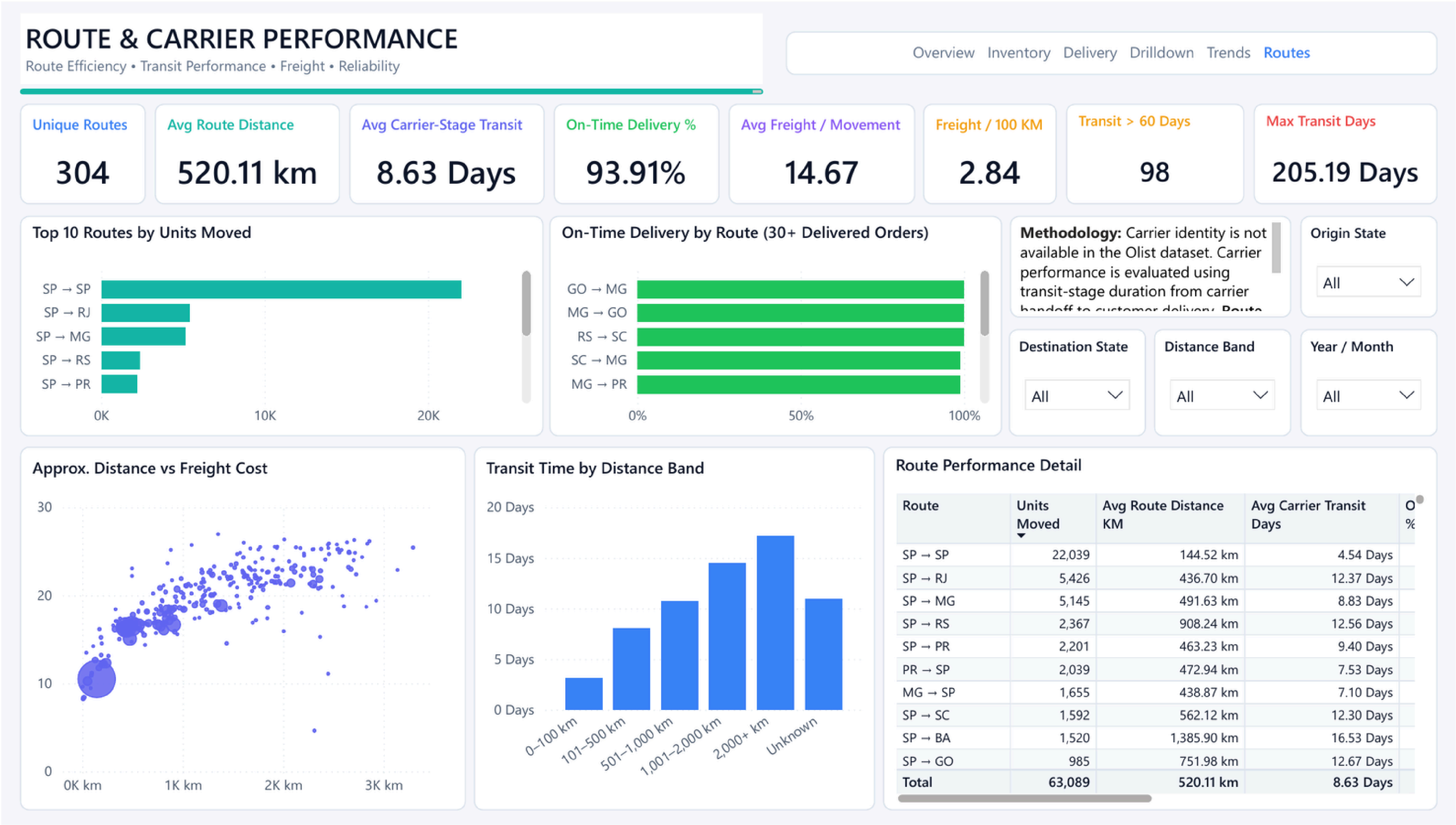
Business interpretation

Prioritise high-cost lanes where service does not improve proportionally with spend.

DASHBOARD 3

Route & Carrier-Stage Performance

A consolidated page compares route volume, approximate distance, transit time, freight efficiency, reliability and extreme exceptions.



Route	Units Moved	Avg Route Distance KM	Avg Carrier Transit Days
SP → SP	22,039	144.52 km	4.54 Days
SP → RJ	5,426	436.70 km	12.37 Days
SP → MG	5,145	491.63 km	8.83 Days
SP → RS	2,367	908.24 km	12.56 Days
SP → PR	2,201	463.23 km	9.40 Days
PR → SP	2,039	472.94 km	7.53 Days
MG → SP	1,655	438.87 km	7.10 Days
SP → SC	1,592	562.12 km	12.30 Days
SP → BA	1,520	1,385.90 km	16.53 Days
SP → GO	985	751.98 km	12.67 Days
Total	63,089	520.11 km	8.63 Days

Unique Routes
304
Seller → customer state

Avg Distance
520.11 km
Approx. geography

Avg Transit
8.63 Days
Carrier-stage duration

On-Time
93.91%
Delivery reliability

Transit > 60 Days
98
Extreme exceptions

Max Transit
205.19 Days
Worst-case outlier

Methodology note

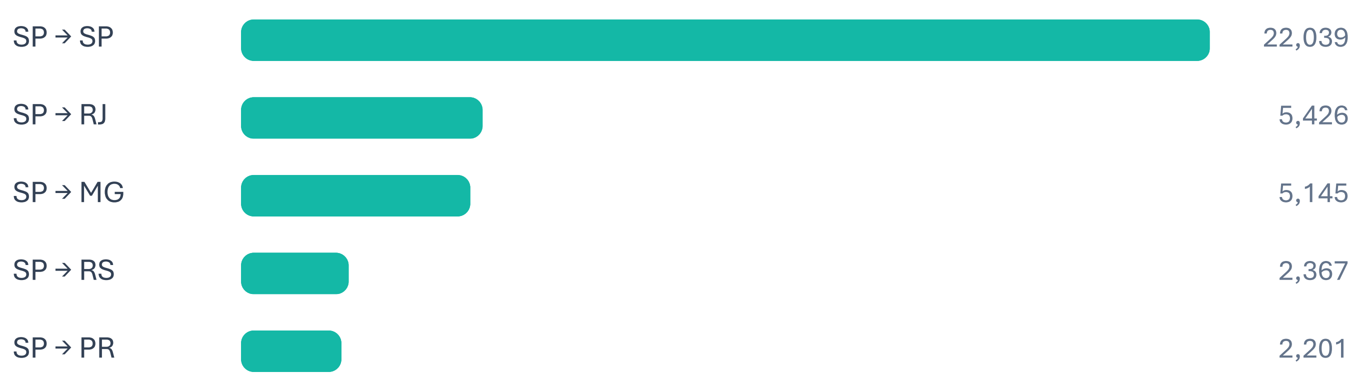
Olist has no carrier identity. “Carrier performance” therefore means transit-stage performance from carrier handoff to customer delivery. Distance is straight-line geography, not road distance.

KEY FINDINGS

Route & Carrier-Stage Performance — Findings & DAX

High-volume lanes, distance effects and exception monitoring turn route data into operational priorities.

Top route volumes



DAX — FREIGHT & TRANSIT EFFICIENCY

```
Total Freight Cost = SUMX(VALUES([Movement Line Key]), MAX([freight_value]))

Freight Cost per 100 KM = DIVIDE([Total Freight Cost], [Total Route Distance KM]) * 100

Avg Carrier Transit Days = AVERAGEX(VALUES([order_id]), MAX([Carrier Transit Days]))
```

1 São Paulo dominates movement

SP → SP alone carries 22,039 units; together with SP → RJ and SP → MG, the top three lanes represent a major share of network movement.

2 Transit generally rises with distance

SP → SP averages 4.54 days at ~145 km, while SP → BA averages 16.53 days at ~1,386 km — supporting distance-specific benchmarks.

3 Exceptions deserve separate management

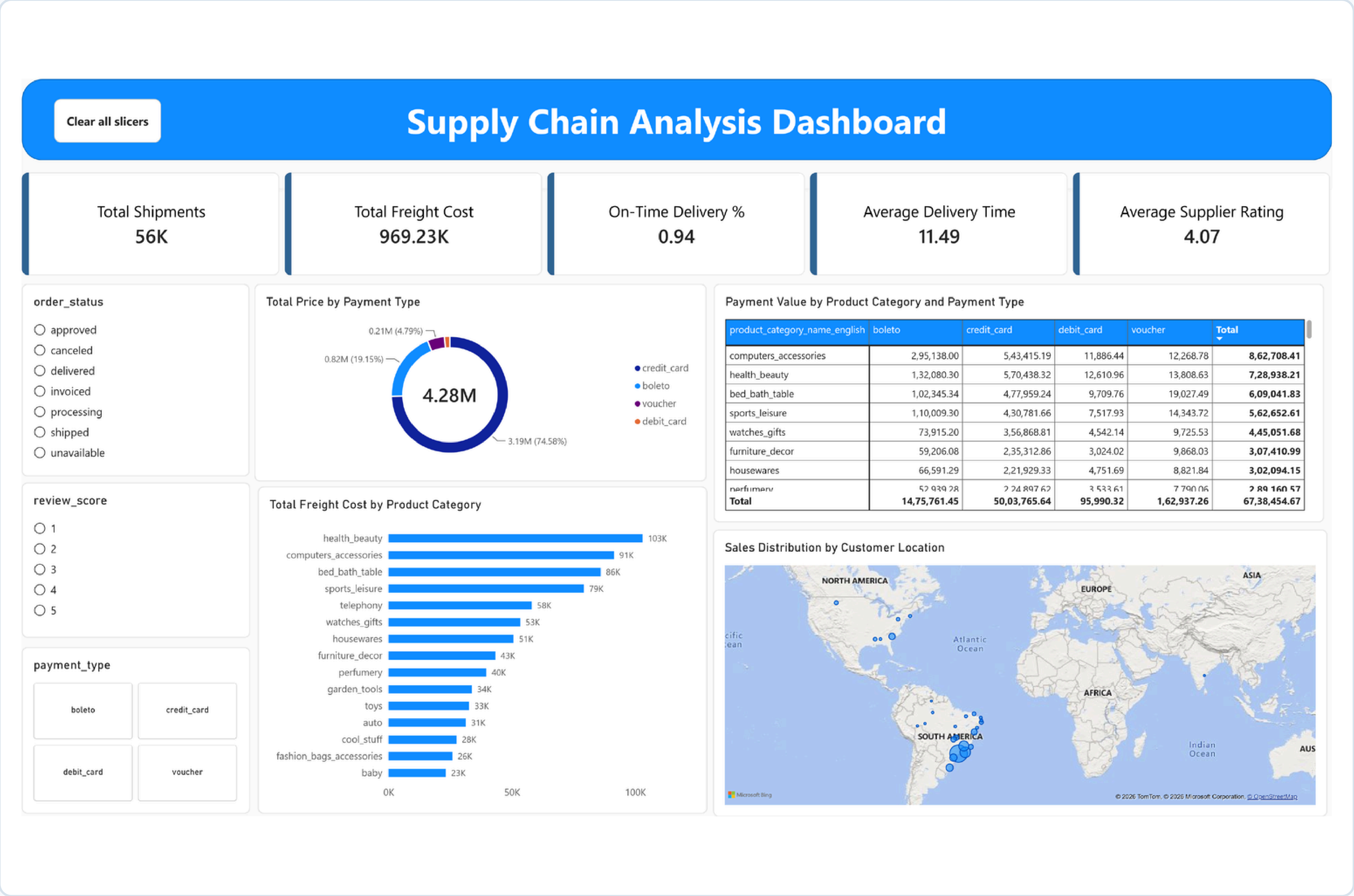
98 orders exceed 60 carrier-stage days and the maximum reaches 205.19 days. These are not representative averages — they are exception cases.

Calculated column logic: Route = seller_state & " → " & customer_state • Approx. distance uses Haversine coordinates (Earth radius 6,371 km).

DASHBOARD 4

Advanced KPI Visualization — Management Overview

A consolidated view combines shipment volume, freight, delivery performance, supplier rating, payment mix and geographic distribution.



Shipments

56K

Dashboard total

Freight Cost

969.23K

Total freight

On-Time

0.94

≈ 94%

Avg Delivery

11.49

Days

Supplier Rating

4.07

Average review

Payment Mix

Credit card dominates

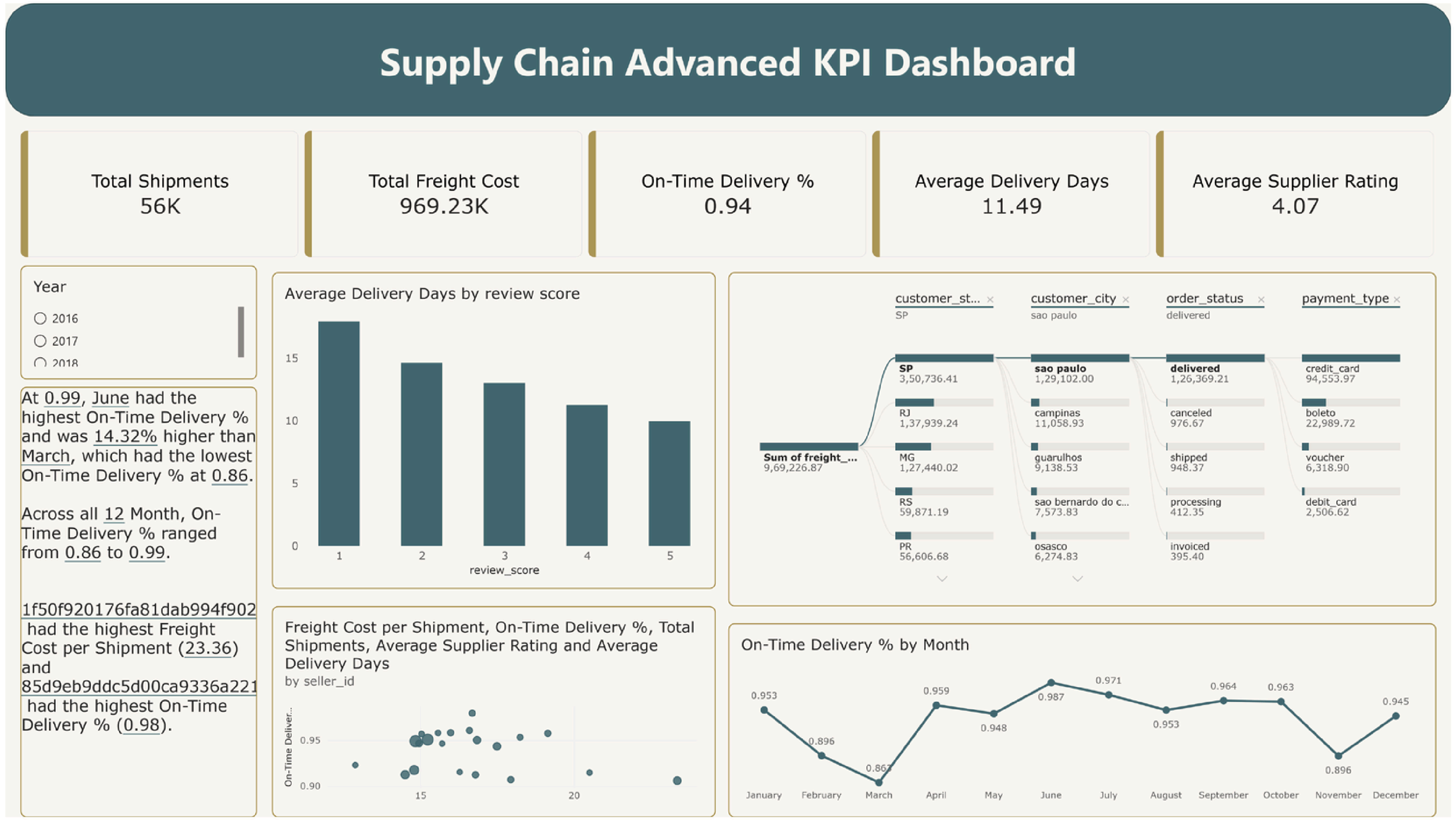
CORE KPI LOGIC

```
On-Time Delivery % =
  On-Time Delivered Orders / Delivered Orders

Average Supplier Rating =
AVERAGE(review_score)
```

Advanced KPI Visualization — Service Pattern Analysis

The second KPI page connects delivery speed, review score, monthly reliability and drillable freight context.



OBSERVED PATTERNS

1 Monthly service varies

The dashboard shows on-time delivery ranging roughly from 0.86 to 0.99 across the twelve displayed months.

2 March vs June

March is the weakest month in the narrative insight, while June is the strongest — a useful target for seasonality/root-cause review.

3 Review & delivery relationship

Average delivery days decline as review score increases in the displayed bar chart — a descriptive association, not proof of causation.

SUPPORTING LOGIC

`Average Delivery Days = mean purchase-to-delivery duration`

CROSS-DASHBOARD VIEW

Integrated Milestone 3 Business Insights

Reading the four workstreams together reveals where the network is strong, where it is costly and where targeted action can create value.

1

Service is strong overall

93.91% on-time delivery provides a solid network baseline — but aggregate reliability can hide local exceptions.

2

Seller improvement is material

109 of 235 eligible sellers are Watchlist or Needs Improvement, making seller governance a major opportunity.

3

Cost is geographically concentrated

Transportation spend is uneven across states and lanes; high-cost areas deserve cost-to-service comparison.

4

Distance shapes transit behavior

Transit time generally rises across route-distance bands, so one global transit target is not operationally fair.

5

Exceptions need a dedicated workflow

98 orders above 60 transit days and a 205-day maximum show why exception monitoring must sit beside average KPIs.

6

Customer experience connects to operations

Higher review scores align with shorter delivery durations in the KPI view, reinforcing the business value of reliable fulfilment.

Management takeaway: optimise the combination of COST + SERVICE + SELLER PERFORMANCE + ROUTE RELIABILITY — not any KPI in isolation.

BUSINESS
RECOMMENDATIONS

Recommended Actions

Translate Milestone 3 analytics into a focused operating rhythm for supplier, transport and delivery improvement.

#	Action	Description	Priority
01	**Supplier governance**	Create monthly scorecard reviews for Watchlist and Needs Improvement sellers; track the weakest component score first.	HIGH
02	**Route exception monitoring**	Flag carrier-stage transit above route/distance benchmarks and prioritise >60-day exceptions for root-cause investigation.	HIGH
03	**Freight optimisation**	Investigate high freight-per-distance lanes and categories where additional spend does not translate into better delivery outcomes.	HIGH
04	**Service benchmark by distance**	Set expected transit ranges by distance band instead of applying a single network-wide target.	MEDIUM
05	**KPI operating cadence**	Use the Advanced KPI view as a monthly management scorecard linking volume, freight, delivery, reviews and geography.	MEDIUM

Suggested cadence: Weekly exceptions • Monthly seller/route review • Quarterly freight optimisation deep dive

ACHIEVEMENT

Milestone 3 Outcome — From Visibility to Performance Management

The project now connects clean data, reliable measures and operational dashboards into a coherent decision-support story.

119,143	→	66,039	→	5	→	4
Raw rows		Cleaned rows		Milestone 2 pages		Milestone 3 workstreams

WHAT MILESTONE 3 ADDS

- A transparent seller scorecard with minimum-volume eligibility and quartile scoring.
- Transportation-cost views that isolate expensive states, lanes, categories and time periods.
- Route performance enriched with approximate geographic distance and carrier-stage transit analysis.
- Advanced KPI pages that link service, cost, ratings, payments and geography for management review.

Final Message: **Milestone 3 transforms the Power BI control tower from descriptive visibility into performance-focused decision support.**

THANK YOU

for your time, guidance and continuous support

Supply Chain Visibility & Optimisation
Milestone 3 — Performance Analytics

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